LESSON 2:

Basic Mathematic Concepts

1. Lines

- A line is a one-dimensional geometric shape that extends infinitely in both directions.
- It can be defined by two points or by an equation.
- There are different types of lines, such as straight lines, curved lines, and segmented lines.
- Equation of a Line (in Slope-Intercept Form): **y=mx+c**, where m is the slope and c is the y-intercept.
- **Slope**: The slope (m) represents the steepness of the line.
- **Parallel Lines:** Lines with the same slope but different y-intercepts are parallel.
- **Perpendicular Lines**: Lines with slopes that multiply to -1 are perpendicular.

Lines Cont'd…

Example: Consider the line with the equation $y=2x+3$.

1.Slope: The slope (m) is 2.

2.Y-intercept: The y-intercept (c) is 3.

3.Plotting the Line:

1. Start by plotting the y-intercept at (0, 3).
2. Use the slope to find another point, say (1, 5) or (-1, 1).
3. Connect the points to draw the line.

2. Curves

- A curve is a one-dimensional geometric shape that does not extend infinitely in both directions.
- It can be defined by an equation or by a set of points.
- There are many different types of curves, such as circles, ellipses, parabolas, and hyperbolas.

1.Types of Curves:

1. **Linear Curve:** Follows a straight line.
2. **Quadratic Curve:** Described by a quadratic equation ($y=ax^2+bx+c$).
3. **Exponential Curve:** Growth or decay described by $y=ab^x$.
4. **Trigonometric Curves:** Such as sine and cosine functions.

2.Circle:

1. Equation of a circle with center (h,k) and radius r: $(x-h)^2+(y-k)^2=r^2$.

3. Matrices

- A matrix is a rectangular array of numbers
- Matrices are used to represent and manipulate data in many different areas of mathematics.
- A matrix can be of any size, but square matrices (matrices with the same number of rows and columns) are the most common type.
- Matrices can be added, subtracted, multiplied, and divided.
- Matrix Operations:
  - Matrix Addition: Add corresponding elements.
  - Scalar Multiplication: Multiply each element by a scalar.
  - Matrix Multiplication: Follow specific rules for combining matrices.
  - Transpose: Swap rows and columns.

Matrices Cont'd…

- Matrix Addition

$$A = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 0 \\ 7 & 3 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 2 & 1 \\ 5 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 7 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 1 \\ 12 & 7 \end{bmatrix}$$

- Scalar Multiplication

Scalar Multiplication

Multiply each entry in the matrix by the given scalar.

Example:

$$A = \begin{bmatrix} 3 & 7 \\ 9 & 10 \end{bmatrix}$$

$$2A = \begin{bmatrix} 2 \cdot 3 & 2 \cdot 7 \\ 2 \cdot 9 & 2 \cdot 10 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & 14 \\ 18 & 20 \end{bmatrix}$$

Matrices Cont'd…

- Matrix Multiplication

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix} = \begin{pmatrix} 70 & 80 & 90 \\ 158 & 184 & 210 \end{pmatrix}$$

- Transpose

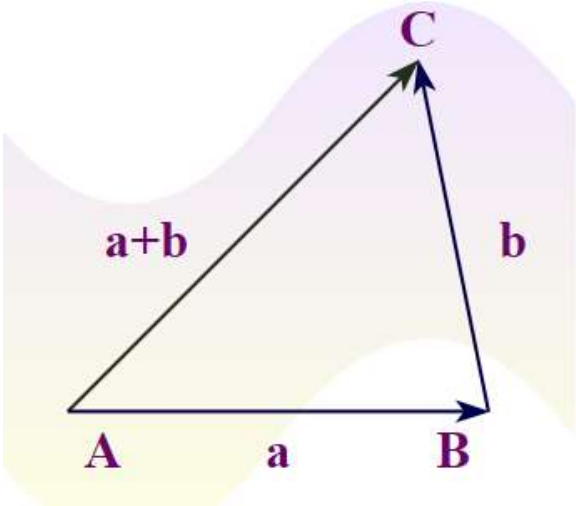
$$\begin{bmatrix} 1 & 4 & 6 \\ 8 & 9 & 7 \\ 5 & 3 & 2 \end{bmatrix}^{T} = \begin{bmatrix} 1 & 8 & 5 \\ 4 & 9 & 3 \\ 6 & 7 & 2 \end{bmatrix}$$

Original Matrix Transpose Matrix

4. Vectors

- A vector is a quantity that has both magnitude and direction.
- Vectors can be represented as ordered pairs (2D), triplets (3D), etc
- Vector operations are:
- Addition
- Subtraction
- Scalar Product
- Cross Product
- Dot Product

Vector Addition



$$\vec{A} = \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix} \qquad \vec{B} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

$\vec{A} + \vec{B}$

Add corresponding elements

$$= \begin{bmatrix} 6 \\ -2 \\ 3 \end{bmatrix}$$

Vector Scalar Product

$$\vec{A} = \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$$

$$\lambda \vec{A} \text{ where } \lambda = 2$$

Multiply each of vector values with 2.

$$= \begin{bmatrix} 8 \\ -6 \\ 2 \end{bmatrix}$$

Vector Cross Product

$$\vec{A} = \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix} \qquad \vec{B} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

$\vec{A} \times \vec{B}$

$$= \begin{bmatrix} 4 & -3 & 1 \\ 2 & 1 & 2 \end{bmatrix}$$

$$= i\begin{bmatrix} -3 & 1 \\ 1 & 2 \end{bmatrix} - j\begin{bmatrix} 4 & 1 \\ 2 & 2 \end{bmatrix} + k\begin{bmatrix} 4 & -3 \\ 2 & 1 \end{bmatrix}$$

$$= i(-6-1) - j(8-2) + k(4--6)$$

$$= -7i - 6j + 10k$$

$$= \begin{bmatrix} -7 \\ -6 \\ 10 \end{bmatrix}$$

Vector Dot Product

$$\vec{A} = \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix} \qquad \vec{B} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix}$$

$$\vec{A} \,.\, \vec{B}$$

Sum of products of transposed vectors corresponding elements

$$= (4 * 2) + (-3 * 1) + (1 * 2) = 8 - 3 + 2 = 7$$